

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide K₄ Fe (CN)₆. After filtering, the precipitate is dried and weighed. The reactions which occur are $\text{ZnS} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2\text{S}$ $2\text{ZnCl}_2 + \text{K}_4\text{Fe}(\text{CN})_6 \rightarrow \text{Zn}_2\text{Fe}(\text{CN})_6 + 4\text{KCl}$ If a 2 gram sample of rock yields 0.969 gram of Zn₂Fe(CN)₆, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight Zn₂ Fe (CN)₆ = 342.6.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

<think>

- We need Zn content in the rock sample , given Zn₂ Fe (CN)₆ content through the stoichi ometric ratios .

- From weight relationships in sto ichi ometry :

Weight Zn₂ Fe (CN)₆ -> Weight Zn = (Weight Zn₂ Fe (CN)₆ * At . Wt Zn) / (Molecular Wt Zn₂ Fe (C N)₆).

- Each 342 . 6 g of Zn₂ Fe (CN)₆ carries 2 * 65 . 4 = 130 . 8 g Zn .

- 0 . 969 g of Zn₂ Fe (CN)₆ -> proportion ate Zn found = (0 . 969 * 130 . 8) / 342 . 6 g .

- The rock sample obtained 2 g ∴ final Ratio = (100 * new -zn) / 2 .

</think>

<answer>

20 % Zn

</answer>